



QR Captain Fleet Management — Product Specification

Document type: Customer-facing feature brief

Audience: Fleet owners, charter operators, commercial fishing operators, marina managers

Purpose: Define the features, roles, and value of the Fleet Management tier before development begins

Status: Draft for stakeholder review

The Problem We're Solving

Running a fleet is fundamentally different from owning a single boat. Right now, if you own five vessels, you have five separate service histories, five separate mechanics, no way to see which boat needs attention first, and no way to know if a missed oil change on your primary charter vessel is about to cost you a cancelled season.

Industry research confirms this pain: **over 30% of marina and fleet operators miss scheduled service appointments** due to manual calendars and disconnected records. Fleet owners with high capital costs per vessel face direct revenue loss when a boat goes down unexpectedly — every cancelled trip, every emergency haul-out, every rush parts order represents money that a better system would have prevented.

QR Captain Fleet Management turns reactive maintenance into proactive asset protection.

The goal is to help people responsible for multiple vessels see maintenance needs, vessel readiness, mechanic activity, service records, and upcoming costs in one place.

This brief presents the features that currently appear most valuable. It is not a final development scope, technical specification, pricing plan, or commitment that every feature listed will be included in the first release.

Stakeholder feedback will be used to:

- Confirm the most important fleet-management problems
- Identify the features that provide the greatest value
- Remove unnecessary complexity
- Separate initial capabilities from future opportunities
- Guide later product and development planning

Who This Is For

Fleet Manager

A Fleet Manager is a vessel owner who operates multiple boats — whether that's a charter operation, a fishing fleet, a racing team, or a family with several vessels at the same marina. They are responsible for keeping the whole fleet earning, not just maintaining a single boat.

Fleet Managers get:

- A dedicated fleet command center that shows the health of every vessel at a glance
- Predictive alerts before problems happen
- One-click mechanic authorization across all vessels at once
- Full financial visibility — cost per boat, upcoming expenses, fleet profitability

Mechanic (Fleet-Tier)

A mechanic assigned to a fleet gains access to every vessel in that fleet through a single authorization. They see service priorities ranked across the fleet so they always know which vessel needs attention most urgently. Their completed work, parts logged, and hours recorded automatically feed the fleet owner's dashboard in real time.

Managing several vessels is different from managing one boat.

Fleet operators may need to track:

- Which vessels are ready to operate
- Which vessels need service first
- Which maintenance items are overdue or approaching
- Which mechanic is responsible for each vessel
- Which work orders are active
- Which costs are expected
- Whether important service and equipment records are complete

When this information is spread across separate records, calendars, calls, text messages, invoices, and paper files, it becomes harder to prevent missed maintenance and avoidable downtime.

The QR Captain Fleet Management concept would bring this information together in a single fleet-level experience.

Individual Boat Owner (Unchanged)

Existing single-vessel owners keep everything they have today. Fleet Management is an upgrade tier for operators managing multiple vessels — it does not change or replace the existing owner experience.

Core Features

The intended experience is simple:

1. Open one fleet dashboard.
2. See which vessels are ready, approaching service, overdue, in maintenance, or out of service.
3. Select any number, alert, or vessel to view the supporting details.
4. Coordinate mechanics and work orders without managing each boat separately.
5. Review maintenance records and expected costs across the fleet.
6. Receive useful alerts before maintenance is missed.

The system should reduce calls, disconnected records, and manual follow-up without overwhelming the user with unnecessary operational detail.

1. Fleet Command Center (Dashboard)

The moment a Fleet Manager logs in, they see the entire fleet — not a list of boats, but a health summary that tells them exactly what needs their attention and what can wait.

What they see at a glance:

- **Fleet Health Score** — A single number (0–100%) representing the percentage of vessels that are fully current on all maintenance. Green means you're running clean. Red means someone has a problem.
- **Vessels In Service / In Maintenance / Out of Service** — Live status for every boat, color-coded: green (operational), yellow (in the shop), red (down).
- **Service Overdue** — Vessels that have already passed their service threshold, ranked by how overdue they are. These need a call today.
- **Service Approaching** — Vessels coming up on their next service window in the next 14 days or 20 engine hours. These need to get on the calendar now.
- **Active Work Orders** — Every open job across the entire fleet, with which boat it's on and who's doing the work.
- **Parts Flagged for Replacement** — Components identified during inspections that are approaching end-of-life, with estimated replacement cost so there are no surprise invoices.

- **Estimated Upcoming Maintenance Cost** — The total projected spend across all open quotes and flagged parts. Know what's coming before it hits your account.
- **Fleet Utilization** — How many hours per week each vessel is running, averaged across the fleet. Identifies which boats are working hard and which are sitting idle.
- **Mechanic Coverage** — Which vessels have an assigned mechanic and which don't. No vessel should be running without someone responsible for its service.
- **Warranty Expiry Watch** — Equipment warranties expiring in the next 90 days, so you can file claims before the window closes.

The dashboard may show:

- Total vessels
- Vessels ready for service
- Vessels currently in maintenance
- Vessels out of service
- Maintenance overdue
- Maintenance approaching
- Active work orders
- Quotes awaiting review
- Flagged parts or inspection findings
- Vessels without an assigned service provider
- Upcoming warranty dates
- Recent fleet activity

Every number on this dashboard is clickable. Tap "3 Vessels Overdue" and you get the three specific boats, ranked by urgency, with their full service status.

2. Predictive Service Scheduling

This is the feature that separates proactive fleet managers from reactive ones.

Instead of relying on a calendar date alone, QR Captain tracks **engine hours** — the actual hours your engines run — and uses them to calculate when each vessel will realistically hit its next service interval. A charter boat that runs 30 hours a week needs service on a completely different timeline than a leisure vessel that goes out twice a month.

How it works for the fleet owner:

- Every time a mechanic checks in on a vessel, they log the current engine hours reading
- QR Captain learns the usage pattern for each engine on each boat
- The system calculates a predicted service date based on actual usage, not a generic schedule
- You see a ranked list: which vessel hits its service interval first, which is furthest away

- As usage picks up (peak season, busy charter weeks), the predictions update automatically

What this prevents:

- Scheduling a service two weeks from now for a boat that will be overdue in four days
 - Discovering mid-season that three vessels all need service at the same time
 - Cancelling a charter because you didn't know an oil change was overdue
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3. Engine Hours Tracking

Engine hours are the odometer of a marine vessel — but unlike a car, there's no dashboard you can glance at from the dock. Fleet managers have historically had to call the marina, call the captain, or wait for the mechanic to report back.

With QR Captain Fleet Management:

- Mechanics log hours directly in the app every time they're aboard
- The full history of every reading is stored per engine, per vessel
- The fleet owner can see current hours, last reading date, and trend (is this boat running more than usual?)
- Engine hour milestones automatically trigger the right maintenance tasks — no manual calendars required

This data feeds directly into the predictive service scheduling engine. The more readings logged over time, the more accurate the predictions become.

Authorized users may record engine-hour readings for each applicable engine or equipment item.

The fleet view may display:

- Current recorded hours
- Date of the last reading
- Reading history
- Usage trend
- Hours remaining before scheduled service
- Maintenance tasks triggered by hour milestones

Client value

Fleet managers gain a clearer view of vessel use and upcoming maintenance needs without calling each captain, marina, or mechanic.

4. Fleet Organization

Fleet Managers can organize their vessels into named fleets. If you run two separate charter operations out of different marinas, they stay separate. If you want to group your racing boats separately from your rental fleet, that's a two-minute setup.

Each fleet has:

- A name, description, and fleet type (charter, fishing, racing, leisure, commercial)
- A roster of vessels — any vessel you own can be assigned to a fleet
- One or more authorized mechanics — when you add a mechanic to a fleet, they get access to every boat in it instantly
- A shared service calendar showing upcoming maintenance across all vessels

Vessels can be moved between fleets or removed from a fleet at any time. Removing a vessel from a fleet doesn't delete its history.

Client value

Organizations with different vessel groups can keep each operation organized while still using one system.

5. One-Click Mechanic Authorization for the Entire Fleet

Today, authorizing a mechanic requires adding them to each vessel individually. For a fleet of 12 boats, that's 12 separate actions.

With Fleet Management, the Fleet Manager authorizes a mechanic **once** for the entire fleet. Every vessel is covered immediately. If you bring on a new boat mid-season, it automatically inherits the fleet mechanic authorization.

When you part ways with a mechanic, revoking their access from the fleet removes them from every vessel in one action.

Client value

The fleet manager can coordinate service access more efficiently while maintaining control over vessel information.

6. Vessel Status Management

Every vessel in the fleet carries a live status:

- **In Service** — Operational, ready to run
- **In Maintenance** — Currently being worked on; do not schedule
- **Out of Service** — Down, needs immediate attention
- **Storage** — Winterized or otherwise out of rotation

Mechanics update vessel status as they start and finish work. Fleet managers see status changes in real time — no more calling to ask "is the Sea Ray ready yet?"

Client value

Fleet managers can see vessel availability without relying on repeated calls or text messages.

7. Fleet Work-Order Visibility

Fleet managers may view active and completed work orders across all authorized vessels.

The fleet work-order view may include:

- Vessel
- Service provider
- Work-order status
- Scheduled date
- Estimated completion date
- Quote status
- Parts status
- Photos and notes
- Completion summary

Client value

Service activity is visible in one place instead of being separated across individual vessel records.

8. Maintenance Cost Visibility

Fleet managers often don't know what a maintenance season is going to cost until the invoices arrive. QR Captain Fleet Management changes that.

- Every part used on a work order is logged with cost
- Parts identified as needing future replacement are flagged during inspections

- The fleet dashboard aggregates all open quotes, pending parts, and flagged replacements into a single **upcoming cost estimate**
- Historical cost per vessel shows which boats are high-maintenance and which are cost-efficient

This gives fleet owners the financial visibility to make smart decisions: when to repair vs. retire a vessel, when to add capacity, and how to price charter trips to actually cover operating costs.

Possible categories include:

- Approved scheduled work
- Quotes awaiting approval
- Completed work
- Flagged future replacements
- Unpriced maintenance needs
- Maintenance cost history by vessel

Client value

Fleet managers can see expected maintenance spending earlier and identify vessels with higher service costs.

9. Digital Vessel Records

Every vessel in the fleet maintains a complete digital profile that travels with the boat — accessible from any device, anytime:

- Service history
- Registration and insurance documents
- Full service history with photos
- Equipment list with serial numbers and warranty dates
- Work order history and invoices
- Engine hours log
- Inspection records
- Invoices and quotes

When you sell a vessel, the complete documented maintenance history is one export away. Documented history increases resale value and protects you in insurance claims.

Client value

Important vessel information remains organized and accessible when needed for service, planning, resale discussions, or insurance documentation.

10. Mobile-First for Mechanics

Mechanics spend their days at the dock, not at a desk. The QR Captain mobile app gives them full access to everything they need from the boat:

- Pull up a vessel's complete service history by scanning the QR code
- Log engine hours, update work order status, and add photos from dockside
- Receive push notifications when a new job is assigned or a part arrives
- See their full fleet assignment list ranked by service urgency — no guessing which boat to hit first
- Log completed work offline; it syncs automatically when connectivity is restored

Possible actions include:

- Scan the vessel's QR code
- Review service history
- Record engine hours
- Start or update a work order
- Add photos and notes
- Flag a part or maintenance concern
- Update maintenance status
- Mark work complete

Client value

Service information can be captured where the work happens instead of being reconstructed later.

11. Notifications and Alerts

The right alert at the right time prevents the problems that cost fleet owners the most money. QR Captain Fleet Management sends smart notifications across web and mobile — no alert fatigue, only actionable signals.

Alert	When It Fires	Why It Matters
Service Overdue	A vessel passes its service interval	Missed PM statistically guarantees higher breakdown frequency within 60–90 days
Service Approaching	20 engine hours or 14 days before service interval	Enough lead time to schedule without disrupting operations
Work Order Complete	Mechanic marks a job finished	Fleet manager knows the vessel is back in service without calling
New Quote Received	Mechanic submits a repair estimate	Approve or decline from your phone
Part Flagged for Replacement	Mechanic identifies a part to watch	Early warning on costs before they become emergencies
Warranty Expiring Soon	Equipment warranty within 90 days of expiry	File claims while the window is still open
Vessel Status Changed	Boat goes in or out of service	Real-time fleet availability without calls or texts
Mechanic Authorization	A mechanic is added or removed from the fleet	Always know who has access to your vessels
Work Order Overdue	A job hasn't been completed by its scheduled date	Prevent billing delays and service slip-through
Engine Hours Milestone	A new hours log reading is submitted	Confirms the mechanic has checked in and triggers service predictions

All notifications are configurable. Fleet managers can choose which alerts come as push notifications, which come as emails, and which are silent dashboard updates only.

Delivery options may include:

- In-app notification
- Email
- Mobile push notification
- Dashboard-only update

Client value

Users receive the information that matters without being overwhelmed by unnecessary alerts.

12. Possible Future Opportunities

These features may be considered later if the core fleet-management experience proves valuable.

- Telematics or engine-data integrations
- Automated engine-hour collection
- Booking or charter-system integrations
- Trip and captain logs
- Advanced maintenance forecasting
- Component-condition monitoring
- GPS-supported service check-in
- Advanced warranty workflows
- Parts inventory management
- Crew scheduling
- Full offline operation
- Multi-location service operations
- Revenue and profitability reporting
- Repair-versus-retire analysis
- Custom reports and data exports

What Fleet Managers Can't Do Without (The Must-Haves)

Based on research across marine fleet operators and charter businesses, these are the capabilities that create the strongest retention — the ones operators say they "can't run the business without" once they have them:

1. Service Urgency Ranking

Not just "which boats need service" but "which boats need service *first*." A ranked list, updated daily, that lets the fleet manager and mechanic prioritize without making judgment calls from incomplete information.

2. Predictive Service Dates

Knowing that Vessel A will hit its service interval in 8 days while Vessel B has 34 days left is the difference between proactive scheduling and reactive scrambling. Once you have this, going back to a paper calendar is not an option.

3. One Number for Fleet Health

A single Fleet Health Score that moves in response to real conditions. When that number is trending down, you know to dig in before something breaks. When it's at 95%, you can take a breath.

4. Cost Forecast Before the Invoice

Seeing upcoming maintenance costs aggregated at the fleet level — before the work is done — is what lets fleet owners make smart business decisions about pricing, timing, and asset deployment.

5. Real-Time Vessel Status

A charter operator cannot schedule a trip on a vessel that's in maintenance. A fishing fleet manager cannot send a captain to a boat that's out of service. The moment a mechanic starts or finishes work, the fleet status updates. No calls, no surprises.

How It Fits Together: A Day in the Life

Fleet Manager — Morning Check (2 minutes)

Opens QR Captain to the Fleet Dashboard. Fleet Health Score is 87%. Two vessels are in the yellow — approaching service. One vessel is in maintenance with an open work order. Clicks the service-approaching alert and sees both boats have oil changes due within the next two weeks. Sends a quick message to the mechanic to confirm they are on the scheduled.

Mechanic — Dockside (throughout the day)

Arrives at the marina. Opens the QR Captain app and sees three vessels ranked by service priority. Scans the QR code on the first vessel, logs current engine hours, starts a work order, takes before photos. Marks the job complete when done. The fleet owner's dashboard updates instantly.

Fleet Manager — Planning for Next Month

Opens the cost forecast. Sees 3,200 in upcoming scheduled maintenance and 800 in flagged parts across the fleet. Adjusts charter pricing for next month to ensure the maintenance budget is covered. No surprises.

Roles Summary

Role	What They See	What They Can Do
Fleet Manager	Fleet dashboard, all vessels, all work orders, cost forecast, service predictions	Create fleets, add vessels, authorize mechanics, approve quotes, view all history
Mechanic (Fleet)	All vessels they are authorized for, ranked by service urgency	Log engine hours, create and update work orders, submit quotes, flag parts, update vessel status
Individual Owner	Their own vessel(s), work orders, their mechanic	Request service, view history, rate mechanics (unchanged from today)

Role	What They See	What They Can Do
Admin	Everything	Full platform access and management

What This Brief Does Not Yet Define

This concept brief does not yet finalize:

- Pricing
- Fleet-size tiers
- Exact user permissions
- Final dashboard layout
- Technical architecture
- Data-storage rules
- Integration requirements
- Development phases
- Release timing
- Mobile offline behavior
- Final forecasting calculations
- Legal or contractual responsibilities
- Final feature commitments

All features will not be available on initial releases so it is also important to prioritize the feature list.

Those decisions should be addressed only after Clance confirms the direction and priorities.

Questions for the Client to Resolve

Before the feature is built, the following decisions need stakeholder input:

1. **Fleet size tiers** — Should Fleet Management be a flat add-on for any number of vessels, or tiered (e.g., up to 10 vessels / 11–25 / 25+)?
2. **Mechanic-initiated fleet setup** — Can a mechanic create a "fleet shell" and invite the boat owner to claim it, or does the Fleet Manager always create the fleet first? (The latter is simpler; the former is more flexible for mechanics who manage clients.)

3. **Crew / captain management** — Should Fleet Managers be able to add captains and crew members with limited access (view vessel status, log engine hours, but not manage work orders)?

4. **Mobile GPS check-in for mechanics** — Should mechanics' log entries auto-capture their GPS location (confirming they were physically at the vessel)? Useful for accountability in commercial operations.